



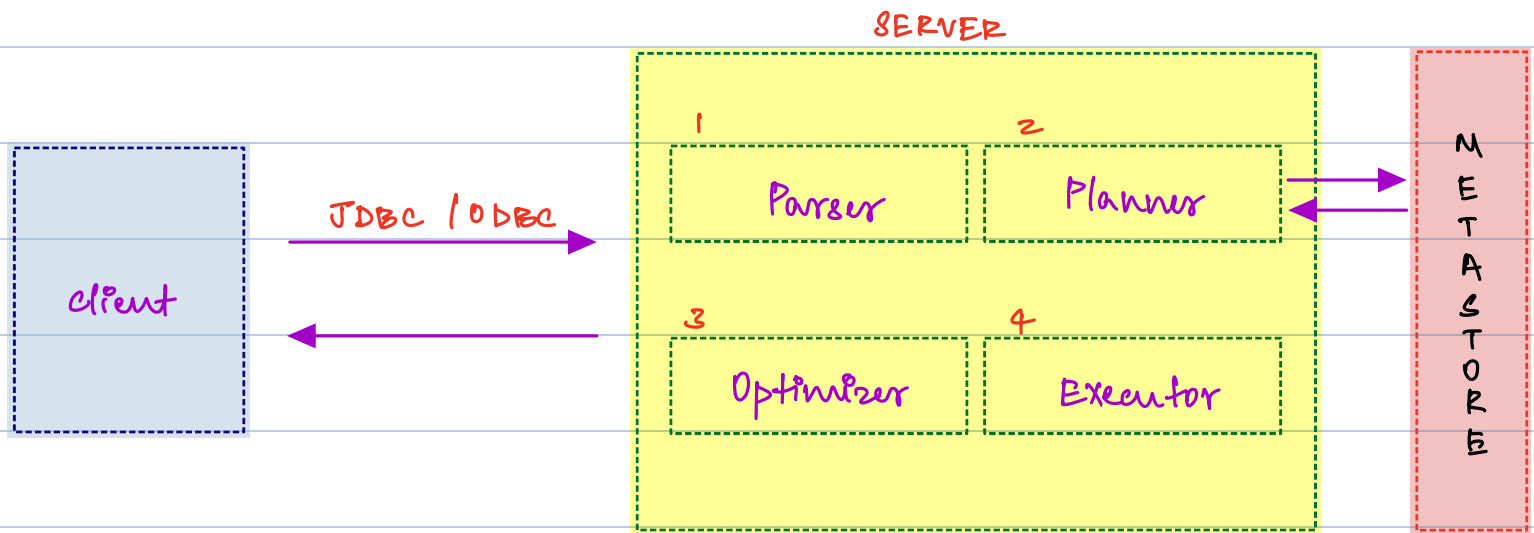
WhatsApp Link

<https://chat.whatsapp.com/EvVhJ555AGW1cJaeKI0USe>

Agenda

- ① Hive Architecture
- ② What happens when you submit a SQL
- ③ Creating a Database and Tables
- ④ Loading the data to table
- ⑤ Optimizing Hive Query
- ⑥ File Format, Partitioning, Better Engine for Execution
- ⑦ Demo

Hive Architecture



Client:

The entry point for the user, this is how users interact with the Data using SQL.

Server: Hive Server 2

Long running Java service, that can handle concurrent.

long running sessions from clients. It executes a series of predefined steps for each of the request.

Parser

The very first step. It checks for any syntactical error in the query and generates an Abstract Syntax Tree to pass it on to Planner.

Planner, Metastore

The planner takes the AST as input. Identifies the tables and columns involved in the query. Communicates with the Metastore to understand location, format etc and prepares stages to implement [logical structure]

Optimizer

The analyser analyses the stages and based on the table statistics it optimises the query using cost based optimization techniques.

Executor

The driver does not execute the plan itself. It uses engines like Map-Reduce, Tez or Spark for executing the logic.

What happens when you submit a SQL query

Client → Driver → Map-Reduce / Tez / Spark

Demo

① Explain Extended

Creating a Database and Table

① Create Database

Hive own data

② Create Managed Table → I need to load data

③ Create External Table → needs a location

User owns the location

Amazon Dataset

Step 1: Create Database

Step 2: create all tables for dimensions and fact table

Step 3: Run Basic Queries.

```
use Database;
```

```
SELECT customername, sum (Revenue)
```

```
from fact join dim Customer
```

```
on customer_id
```

```
Group By 1;
```

Load data into tables

Two types of table in Hive

Majority of us use External table:

Load data	Managed Table	External Table
It moves	Data owns	Data you own data
from src	Metadata owns	Metadata own
to dest		

Optimizing Hive Query

- ① File Format
- ② Partition
- ③ Other Execution Engines

File Format → product [CSV]

ID	name	category	price
1	earphone	electronic	2000
2	phone	electronic	20000
3	laptop	electronic	50000

Pro: You can understand and see the data

Cons:

- ① Storage. compression is low
- ② Not optimized for programs

Apache Parquet

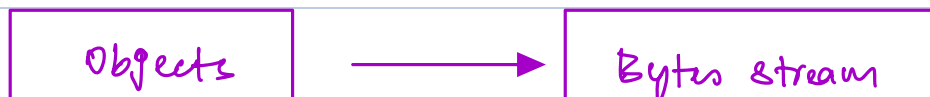
- ① columnar storage

store one single column values together

1	2	3	4
---	---	---	---

electronic	electronic	elec	elec
------------	------------	------	------

- ① provides better compression of data
- ② Helps in serializing & deserializing data



In production → parquet

Apache Avro

Is similar to your JSON structure

Stores key and value

V.V. Pump:

It initializes the fields along with their data types
in the beginning of file

Doubt Resolution

SQL, HDFS, YARN, MR, DWH, Hive

Mondays

- ① Partitions & Buckets
- ② Different Engines

Amazon Books

Revenue from Book sales

① ^{Dim} Product

② Fact

4 cols

Product ID

xy2123

Product Na

abc 456.

table	col	location
prod	ID	xy2123
prod	name	abc 456

Derby DB

Oracle